

Theoretical Physics for Mathematicians

Home Work 2

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Question 1 - ★ Hamilton's Principle: The Euler–Lagrange equations and their invariances

Let $L = L(q, \dot{q}; t)$ be smooth and let

$$S[q] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t); t) dt.$$

(a) **(Fundamental lemma).** Let $F : [t_0, t_1] \rightarrow \mathbb{R}^f$ be continuous and suppose

$$\int_{t_0}^{t_1} F_a(t) \eta^a(t) dt = 0$$

for every smooth η with compact support in (t_0, t_1) . Prove that $F = 0$. *Hint: if $F_a(t_*) \neq 0$ for some t_* and some index a , construct a bump function concentrated near t_* .*

(b) Compute the first variation

$$\delta S[q](\eta) = \left. \frac{d}{d\epsilon} S[q + \epsilon \eta] \right|_{\epsilon=0}$$

for variations with $\eta(t_0) = \eta(t_1) = 0$, and deduce from (a) that $\delta S = 0$ is equivalent to the Euler–Lagrange equations.

(c) Let $F = F(q, t)$ be smooth and set

$$L' = L + \frac{d}{dt} F(q, t), \quad \text{i.e.} \quad L' = L + \frac{\partial F}{\partial q^a} \dot{q}^a + \frac{\partial F}{\partial t}.$$

Show that L and L' have exactly the same Euler–Lagrange equations. How do the actions S and S' differ?

(d) **(Coordinate invariance).** Let $q = q(Q)$ be a diffeomorphic change of generalized coordinates and let

$$\tilde{L}(Q, \dot{Q}; t) = L\left(q(Q), \frac{\partial q}{\partial Q} \dot{Q}; t\right).$$

Prove the identity

$$\frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{Q}^b} - \frac{\partial \tilde{L}}{\partial Q^b} = \frac{\partial q^a}{\partial Q^b} \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^a} - \frac{\partial L}{\partial q^a} \right),$$

and conclude that the Euler–Lagrange equations are coordinate independent. Why does the Cartesian component form $m\ddot{x}^a = F^a$ acquire connection terms in curvilinear coordinates?

Answer:

(a) Let us assume that $F_a(t_*) > 0$ for some t_* - it makes no difference if it would be negative in the argument. As F is continuous there is an interval $[t_* - \delta, t_* + \delta]$ for which $F(t) > 0, \forall t \in [t_* - \delta, t_* + \delta]$. Now consider the C^1 function given by:

$$\eta^a(t) = 1 + \sin \left[-\pi/2 + \pi \frac{(t - t_* + \delta)}{\delta} \right], \quad t \in [t_* - \delta, t_* + \delta]$$

and zero elsewhere. Then $\eta^a(t) > 0$ on the same interval that $F_a(t) > 0$ and zero elsewhere - hence their product is greater than zero on $[t_* - \delta, t_* + \delta]$ and zero elsewhere. Hence, for:

$$\int_{t_0}^{t_1} F_a(t) \eta^a(t) dt = 0 \quad \text{for every smooth } \eta^a \text{ with compact support}$$

we must conclude that $F_a(t) = 0$ on (t_0, t_1)

(b)

$$S[q] := \int_{t_0}^{t_1} L(q, \dot{q}, t) dt$$

Let η be a perturbation of q s.t. $\eta(t_0) = \eta(t_1) = 0$.

Then:

$$\begin{aligned} S[q + \epsilon \eta] &= \int_{t_0}^{t_1} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) dt, \\ \Rightarrow \quad \frac{dS}{d\epsilon} &= \int_{t_0}^{t_1} \frac{d}{d\epsilon} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) dt. \end{aligned} \quad (*)$$

Now

$$\frac{d}{d\epsilon} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) = \frac{\partial}{\partial q} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) \eta + \frac{\partial}{\partial \dot{q}} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) \dot{\eta}.$$

So, (*) becomes:

$$\begin{aligned} \frac{dS}{d\epsilon} &= \int_{t_0}^{t_1} \left(\frac{\partial}{\partial q} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) \eta + \frac{\partial}{\partial \dot{q}} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) \dot{\eta} \right) \\ \Rightarrow \quad \frac{dS}{d\epsilon} \Big|_{\epsilon=0} &= \int_{t_0}^{t_1} \left(\frac{\partial}{\partial q} L(q, \dot{q}, t) \eta + \frac{\partial}{\partial \dot{q}} L(q, \dot{q}, t) \dot{\eta} \right) \end{aligned} \quad (**)$$

Integrating by parts:

$$\int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}} \dot{\eta} dt = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta dt.$$

Now $\left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_0}^{t_1} = 0$, since: $\eta(t_0) = \eta(t_1) = 0$. So, (**) becomes:

$$\begin{aligned} \frac{dS}{d\epsilon} \Big|_{\epsilon=0} &= \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q} \eta - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta \right) dt \\ &= \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) \eta dt. \end{aligned}$$

Now, by assumption:

$$\frac{dS}{d\epsilon} \Big|_{\epsilon=0} = \delta S = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) \eta dt = 0.$$

Now η is an arbitrary “perturbation” – so use part (a) of this question to conclude that

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0.$$

□

(c) We are given $F = F(q, t)$ such that:

$$\begin{aligned} L' &= L + \frac{\partial F}{\partial q^a} \dot{q}^a + \frac{\partial F}{\partial t} \\ \Rightarrow \quad \frac{\partial L'}{\partial q^b} &= \frac{\partial L}{\partial q^b} + \frac{\partial^2 F}{\partial q^a \partial q^b} \dot{q}^a + \frac{\partial^2 F}{\partial q^b \partial t}. \end{aligned}$$

We also have that:

$$\begin{aligned} \frac{\partial L'}{\partial \dot{q}^b} &= \frac{\partial L}{\partial \dot{q}^b} + \frac{\partial F}{\partial q^b} \\ \Rightarrow \quad \frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{q}^b} \right) &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^b} \right) + \frac{d}{dt} \left(\frac{\partial F}{\partial q^b} \right) \end{aligned} \quad (***)$$

For the last term in (***):

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial F}{\partial q^b} \right) &= \frac{\partial^2 F}{\partial q^b \partial q^c} \dot{q}^c + \frac{\partial}{\partial t} \left(\frac{\partial F}{\partial q^b} \right) \\ &= \frac{\partial^2 F}{\partial q^b \partial q^c} \dot{q}^c + \frac{\partial^2 F}{\partial t \partial q^b} \end{aligned}$$

Now the derivative order of the 2nd order partials is not relevant (as F is smooth)

and hence:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{q}^a} \right) &= \boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^a} \right)} + \frac{\partial^2 F}{\partial q^a \partial q^b} \dot{q}^b + \frac{\partial^2 F}{\partial t \partial q^a} \\
&= \boxed{\frac{\partial L}{\partial q^a}} + \frac{\partial^2 F}{\partial q^a \partial q^b} \dot{q}^b + \frac{\partial^2 F}{\partial q^a \partial t} \\
&= \frac{\partial L'}{\partial q^a}
\end{aligned}$$

The middle line follows as L satisfies the Langrange equation and the two terms in red boxes are equal. **NOTE: In Einstein notation the double indices are dummy.**

If we consider the action for these two Langrangians for the same path (q):

$$\begin{aligned}
S[q] &= \int_{t_0}^{t_1} L(q, \dot{q}, t) dt \\
S'[q] &= \int_{t_0}^{t_1} L(q, \dot{q}, t) + \frac{dF}{dt} dt \\
&= \int_{t_0}^{t_1} L(q, \dot{q}, t) dt + F(q(t_1), t_1) - F(q(t_0), t_0) \\
&= S[q] + F(q(t_1), t_1) - F(q(t_0), t_0)
\end{aligned}$$

(d) Given $q = q(Q)$ – no time dependence. Then

$$\frac{\partial \hat{L}}{\partial Q^b} = \frac{\partial L}{\partial q^a} \cdot \frac{\partial q^a}{\partial Q^b} \quad (\text{A})$$

Also:

$$\frac{\partial \hat{L}}{\partial \dot{Q}^b} = \frac{\partial L}{\partial \dot{q}^a} \cdot \frac{\partial \dot{q}^a}{\partial \dot{Q}^b}. \quad (\text{B})$$

Now

$$\begin{aligned}
\dot{q}^a &= \frac{\partial q^a}{\partial Q^b} \dot{Q}^b \\
\Rightarrow \frac{\partial \dot{q}^a}{\partial \dot{Q}^b} &= \frac{\partial q^a}{\partial Q^b}. \quad (\text{C})
\end{aligned}$$

Using this equality equation (B) becomes:

$$\frac{\partial \hat{L}}{\partial \dot{Q}^b} = \frac{\partial L}{\partial \dot{q}^a} \frac{\partial q^a}{\partial Q^b} \quad (\text{D})$$

Then, using the product rule, in (D):

$$\frac{d}{dt} \left(\frac{\partial \hat{L}}{\partial \dot{Q}^b} \right) = \left(\frac{\partial q^a}{\partial Q^b} \right) \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^a} \right) + \left(\frac{\partial L}{\partial \dot{q}^a} \right) \left(\frac{d}{dt} \frac{\partial q^a}{\partial Q^b} \right). \quad (\text{E})$$

The last term in equation (E) is time independent - and is hence zero. So (E) becomes:

$$\frac{d}{dt} \left(\frac{\partial \hat{L}}{\partial \dot{Q}^b} \right) = \left(\frac{\partial q^a}{\partial Q^b} \right) \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^a} \right). \quad (\text{F})$$

Lagrange's eqn in terms of $\hat{L}$ is, using (A) and (F):

$$\begin{aligned} \left(\frac{\partial \hat{L}}{\partial Q} \right) - \frac{d}{dt} \left(\frac{\partial \hat{L}}{\partial \dot{Q}^b} \right) &= \left(\frac{\partial L}{\partial q^a} \right) \left(\frac{\partial q^a}{\partial Q^b} \right) - \left(\frac{\partial q^a}{\partial Q^b} \right) \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^a} \right) \\ &= \left(\frac{\partial q^a}{\partial Q^b} \right) \left[\frac{\partial L}{\partial q^a} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^a} \right) \right]. \end{aligned}$$

□

I am not sure what is requested in the last part of this question.

Question 2 - ★ Oscillator and pendulum, with and without multipliers

(a) For

$$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$$

derive the equation of motion and its general solution, compute

$$E_L = \dot{q} \frac{\partial L}{\partial \dot{q}} - L$$

and verify directly that $\dot{E}_L = 0$ on solutions of the Euler–Lagrange equation.

(b) For the plane pendulum of length ℓ use the angle θ (measured from the downward vertical) to derive

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0.$$

Determine the small-oscillation frequency, and show that energy conservation reduces the problem to the quadrature

$$\dot{\theta}^2 = \frac{2g}{\ell} (\cos \theta - \cos \theta_0),$$

where $0 < \theta_0 < \pi$ is the amplitude of a libration. Deduce the exact period as an integral and check that it tends to $2\pi\sqrt{\ell/g}$ as $\theta_0 \rightarrow 0$.

(c) Treat the same pendulum in $\mathbb{R}^2$ with the constraint

$$G(x, y) = x^2 + y^2 - \ell^2 = 0$$

and

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy.$$

Write down the multiplier equations, and determine λ explicitly as a function of $(x, y, \dot{x}, \dot{y})$.

(d) Write the constraint force as

$$\vec{F}_{\text{constraint}} = -\mathcal{T} \hat{e}_r,$$

where

$$\hat{e}_r = (\sin \theta, -\cos \theta)$$

points outwards. Show that the signed inward tension is

$$\mathcal{T} = m \left(g \cos \theta + \ell \dot{\theta}^2 \right), \quad \left| \vec{F}_{\text{constraint}} \right| = |\mathcal{T}|.$$

Explain the distinction between a rigid rod, which can exert compression, and a taut string. Confirm that (c) reproduces the equation of motion of (b).

Answer:

(a)

$$\begin{aligned}
 L &= \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2 \\
 \Rightarrow \frac{\partial L}{\partial \dot{q}} &= m\dot{q} \\
 \Rightarrow E_L &= \dot{q} \frac{\partial L}{\partial \dot{q}} - L \\
 &= \dot{q}(m\dot{q}) - \left[\frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2 \right] \\
 &= \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2
 \end{aligned}$$

But this is The total energy of The System - which for an isolated system must be constant. Indeed:

$$\begin{aligned}
 \dot{E}_L &= m\dot{q}\ddot{q} + kq\dot{q} \\
 &= \dot{q}[m\ddot{q} + kq] \\
 &= 0 \quad \quad \quad (\text{from E-L equations})
 \end{aligned}$$

(b) For The pendulum we have:

$$\begin{aligned}
 \text{k.e. of pendulum} &= \frac{1}{2}mv^2 = \frac{1}{2}ml^2\dot{\theta}^2 \\
 \text{p.e. of pendulum} &= mlg(1 - \cos \theta)
 \end{aligned}$$

Here: p.e = 0 at bottom of pendulum swing.

Only 1 degree of freedom: θ

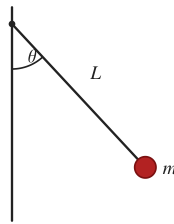


Figure 1: Simple Pendulum

The Lagrangian is:

$$L = \frac{1}{2}ml^2\dot{\theta}^2 - mlg(1 - \cos \theta)$$

Lagrange's dynamical eqn is then:

$$\begin{aligned}
\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} &= \frac{\partial L}{\partial \theta} \\
\Rightarrow \frac{d}{dt} (ml^2 \dot{\theta}) &= -mgl \sin \theta \\
\Rightarrow ml^2 \ddot{\theta} &= -mgl \sin \theta \\
\Rightarrow \ddot{\theta} + \frac{g}{l} \sin \theta &= 0
\end{aligned} \tag{2.1}$$

For **small oscillations** θ is small $\Rightarrow \sin(\theta) \approx \theta$, then (2.1) becomes:

$$\ddot{\theta} = -\frac{g}{l} \theta \tag{2.2}$$

Under such motion, the frequency is given by: $\omega = \sqrt{\frac{g}{l}}$.

For energy conservation - given by E_L , we have that:

$$\begin{aligned}
E_L &= \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} - L \\
&= \dot{\theta} [ml^2 \dot{\theta}] - \left[\frac{1}{2} ml^2 \dot{\theta}^2 - mgl(1 - \cos \theta) \right] \\
&= \frac{1}{2} ml^2 \dot{\theta}^2 + mgl(1 - \cos \theta).
\end{aligned} \tag{2.3}$$

We know that at $\theta = \theta_0$ (max amplitude) all energy is potential energy (hence $\dot{\theta} = 0$). Hence (2.3) becomes

$$\begin{aligned}
mgl(1 - \cos \theta_0) &= \frac{1}{2} ml^2 \dot{\theta}^2 + mgl(1 - \cos \theta) \\
\Rightarrow \frac{1}{2} ml^2 \dot{\theta}^2 &= mgl(\cos \theta - \cos \theta_0) \\
\Rightarrow \dot{\theta}^2 &= \frac{2g}{l} (\cos \theta - \cos \theta_0)
\end{aligned} \tag{2.4}$$

Here we consider determining the period of motion from (2.4).

$$\begin{aligned}
\dot{\theta} &= \sqrt{\frac{2g}{l} (\cos \theta - \cos \theta_0)} \\
\Rightarrow \sqrt{\frac{l}{2g}} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}} &= dt \\
\Rightarrow \sqrt{\frac{l}{2g}} \int_{-\theta_0}^{\theta_0} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}} &= \int_0^{\frac{T}{2}} dt
\end{aligned} \tag{2.5}$$

Where in (2.5) I am integrating over 1/2 period of oscillation, $T/2$, when the pendulum swings from $-\theta_0$ to θ_0 .

Next, for very small θ, θ_0 , expand $\cos(\theta) \approx 1 - \theta^2/2$ and same for $\cos(\theta_0)$. With this equation (2.5) becomes:

$$\begin{aligned} \sqrt{\frac{l}{2g}} \int_{-\theta_0}^{\theta_0} \frac{d\theta}{\left(\frac{1}{\sqrt{2}}(\theta_0^2 - \theta^2)\right)^{1/2}} &= \frac{T}{2} \\ \Rightarrow \sqrt{\frac{l}{g}} \int_{-\theta_0}^{\theta_0} \frac{d\theta}{(\theta_0^2 - \theta^2)^{1/2}} &= \frac{T}{2} \end{aligned} \quad (2.6)$$

The integral in (2.6) is a standard integral and evaluates to $\left[\arcsin\left(\frac{\theta}{\theta_0}\right)\right]_{-\theta_0}^{\theta_0} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$. Substituting this into (2.6) gives:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

(c) To be honest I looked at a video online on how to solve this using Lagrangian Multipliers: **Video Link**.

We can construct a modified Lagrangian, K , that contains both the "usual" Lagrangian, L , together with the constraint, G , and a Lagrangian multiplier, $\lambda(t)$:

$$\begin{aligned} K(x, y, \dot{x}, \dot{y}, t) &= L(x, y, \dot{x}, \dot{y}, t) + \lambda G(x, y) \\ &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy + \lambda(x^2 + y^2 - l^2) \end{aligned}$$

We then perform the usual differentiation for both x and y coordinates as we would have done for the "usual" Lagrangian:

$$\frac{d}{dt} \frac{\partial K}{\partial \dot{x}} = \frac{\partial K}{\partial x} \quad (2.7)$$

$$\frac{d}{dt} \frac{\partial K}{\partial \dot{y}} = \frac{\partial K}{\partial y} \quad (2.8)$$

And we have the additional equation - from the constraint, that:

$$x^2 + y^2 - l^2 = 0 \quad (2.9)$$

Evaluating (2.7) and (2.8) we get:

$$m\ddot{x} = 2\lambda x \quad (2.10)$$

$$m\ddot{y} = -mg + 2\lambda y \quad (2.11)$$

From the constraint we have:

$$\begin{aligned} x^2 + y^2 &= l^2 \\ \Rightarrow 2x\dot{x} + 2y\dot{y} &= 0 \\ \Rightarrow x\ddot{x} + \dot{x}^2 + y\ddot{y} + \dot{y}^2 &= 0 \\ \Rightarrow x\ddot{x} + y\ddot{y} &= -(\dot{x}^2 + \dot{y}^2) \end{aligned}$$

Using (2.10), (2.11) and $x^2 + y^2 = l^2$:

$$\lambda = \frac{m}{2l^2}[gy - (x^2 + y^2)] \quad (2.12)$$

(d)